5.6 Astrophysics and Cosmology

This topic covers gravitational fields and the physical interpretation of astronomical observations, the formation and evolution of stars and the history and future of the universe.

Candidates will be assessed on their ability to:

154	understand that a gravitational field (force field) is defined as a region where a mass experiences a force
155	understand that gravitational field strength is defined as $g = \frac{F}{m}$ and be able to use this equation
156	be able to use the equation $F = \frac{Gm_1m_2}{r^2}$ (Newton's law of universal gravitation)
157	be able to derive and use the equation $g = \frac{Gm}{r^2}$ for the gravitational field due to a point mass
158	be able to use the equation $V_{grav} = \frac{-Gm}{r}$ for a radial gravitational field
159	be able to compare electric fields with gravitational fields
160	be able to apply Newton's laws of motion and universal gravitation to orbital motion
161	understand what is meant by a *black body radiator* and be able to interpret radiation curves for such a radiator
162	be able to use the Stefan-Boltzmann law equation $L = \sigma AT^4$ for black body radiators
163	be able to use Wien's law equation $\lambda_{max}T$ = 2.898 x 10^{-3} m K for black body radiators
164	be able to use the equation, intensity $I = \frac{L}{4\pi d^2}$ where L is luminosity and d is distance from the source
165	understand how astronomical distances can be determined using trigonometric parallax
166	understand how astronomical distances can be determined using measurements of intensity received from standard candles (objects of known luminosity)
167	be able to sketch and interpret a simple Hertzsprung-Russell diagram that relates stellar luminosity to surface temperature
168	understand how to relate the Hertzsprung-Russell diagram to the life cycle of stars
169	understand how the movement of a source of waves relative to an observer/detector gives rise to a shift in frequency (Doppler effect)

170	be able to use the equations for redshift $z = \frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for a source of electromagnetic radiation moving relative to an observer and $v = H_0 d$ for objects at cosmological distances
171	understand the controversy over the age and ultimate fate of the universe associated with the value of the Hubble constant and the possible existence of dark matter.